Practice Assignment 6 - Not Graded

This assignment will not be graded and is only for practice.

Multiple Choice Questions (MCQ):

1 point

Which of the following statements are correct?

- [ ] There exists a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2)$, $T(1,-1) = (1,-1)$, and $T(4,1) = (1,0)$.
- [ ] If there is a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2)$, and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.
- [ ] Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V. Consider a linear transformation $T: V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. The matrix representation of T, with respect to β and γ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$
- [ ] Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W. Consider a linear transformation $T: V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. Then, the rank of T is 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If there is a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2)$, and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.

Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V. Consider a linear transformation $T: V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. The matrix representation of T, with respect to β and γ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$

Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W. Consider a linear transformation $T: V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. Then, the rank of T is 3.

1 point

Consider the following set $S = \{(x, y) \mid x + y = 1, x, y \in R\}$. In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set $T(S)$ given in column B and the geometric representations of S and $T(S)$ in column C.

- [ ] i $\rightarrow$ b $\rightarrow$ 2.
- [ ] ii $\rightarrow$ c $\rightarrow$ 1.
- [ ] ii $\rightarrow$ c $\rightarrow$ 2.
- [ ] iii $\rightarrow$ a $\rightarrow$ 3.
- [ ] i $\rightarrow$ b $\rightarrow$ 3.
- [ ]

iii $\to$ $\to$ a $\to$ $\to$ 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

ii $\to$ $\to$ c $\to$ $\to$ 2.
i $\to$ $\to$ b $\to$ $\to$ 3.
iii $\to$ $\to$ a $\to$ $\to$ 1.

Multiple Select Questions (MSQ):

1 point

Consider the following two groups. In Group 1, there are 3 functions from one vector space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication.

Group-1

- **P:** $T: R^2 \to R^2$ **such that** $T(x, y) = \begin{cases} \left(\frac{x^2}{y}, y\right) & \text{if } y \neq 0 \\ (x, 0) & \text{otherwise} \end{cases}$
- **Q:** $T: R^2 \to R^2$ **such that** $T(x, y) = (x, xy)$
- **R:** $T: R^3 \to R$ **such that** $T(x, y, z) = \frac{x+y+z}{5}$

Group-2

- 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$
- 2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in R$.
- 3: T is a linear transformation

Choose the set of correct options.

- [ ] P satisfies 1 but not 2.
- [ ] P satisfies 2 but not 1
- [ ] Q satisfies 1
- [ ] Q satisfies 3
- [ ] P satisfies 3
- [ ] R satisfies 2
- [ ] R satisfies 3
- [ ] R satisfies 1 but not 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

P satisfies 2 but not 1
R satisfies 2
R satisfies 3

Numerical Answer Type (NAT):

Suppose the matrix representation of a linear transformation $T: R^3 \to R^3$ with respect to the ordered bases $\beta = \{(1,0,0), (1,1,0), (1,1,1)\}$ for the domain and $\gamma = \{(1,1,1), (0,1,1), (0,0,1)\}$ for the range, is $I_{3\times 3}$, i.e., the identity matrix of order 3. Let A be the matrix representation of the linear transformation T with respect to standard ordered basis of $\mathbb{R}^3$ for both domain and range. Then the determinant of A is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

Let T and S be two linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$, which are defined as follows:

$$T : \mathbb{R}^3 \to \mathbb{R}^3 \qquad S : \mathbb{R}^3 \to \mathbb{R}^3$$

$$T(x, y, z) = (x - y, y - z, z - x)$$
$$S(x, y, z) = (x + y, y + z, z + x)$$

Let $S + T$ be defined to be the linear transformation as follows:

$$(S + T) : \mathbb{R}^3 \to \mathbb{R}^3$$

$$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z)$$

Let C be the matrix representation of $S + T$ with respect to the standard ordered bases of $\mathbb{R}^3$. If $C = nI$, where I denotes the identity matrix of order 3, then what will be the value of n?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

1 point

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined as $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$. Which of the following options are true about T?

- [ ] Nullity of T is 0.
- [ ] A basis for the range of T is $\{(1, 5, 0), (0, 1, 6)\}$.
- [ ] A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
- [ ] Rank of T is 2.
- [ ] The matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

☐

The matrix representation of T with respect to the standard ordered bases of R^3 is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

☐

Nullity of T is 1.

☐

Rank of T is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity of T is 0.

A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

The matrix representation of T with respect to the standard ordered bases of R^3 is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

Rank of T is 3.

Let V be a vector space with ordered basis $\alpha = \{v_1, v_2, \ldots, v_{10}\}$. Consider a linear transformation $T: V \to V$ such that $T(v_1) = v_1$ and $T(v_i) = v_i - v_{i-1}$, where $i > 1$. Let A be the matrix representation of the linear transformation T, with respect to α for both the domain and codomain. what is the trace of A?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset S of R^3 consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation $T: R^3 \to R$ to S where R^3 comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored). The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

Use the above information to answer questions 8, 9 and 10.

1 point

Which of the following sets of vectors in SS have the property that the YouTube video corresponding to every linear combination of that set which is in SS has ranking $0$0?

- [ ] $\{(1,1,0),(0,0,1)\}$$\{(1,1,0),(0,0,1)\}$
- [ ] $\{(6,25,0),(0,0,25)\}$$\{(6,25,0),(0,0,25)\}$
- [ ] $\{(25,6,0),(0,0,25)\}$$\{(25,6,0),(0,0,25)\}$
- [ ] $\{(12,50,0),(0,0,1)\}$$\{(12,50,0),(0,0,1)\}$
- [ ] $\{(24,100,0),(0,0,25)\}$$\{(24,100,0),(0,0,25)\}$
- [ ] $\{(6,0,0),(0,25,0),(0,0,1)\}$$\{(6,0,0),(0,25,0),(0,0,1)\}$
- [ ] $\{(25,0,0),(0,6,0),(0,0,1)\}$$\{(25,0,0),(0,6,0),(0,0,1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(6,25,0),(0,0,25)\}$$\{(6,25,0),(0,0,25)\}$

$\{(12,50,0),(0,0,1)\}$$\{(12,50,0),(0,0,1)\}$

$\{(24,100,0),(0,0,25)\}$$\{(24,100,0),(0,0,25)\}$

1 point

Which of the following sets of vectors in SS have the property that the YouTube video corresponding to every linear combination of that set which is in SS has ranking a multiple of $15$15?

- [ ] $\{(3,0,0),(0,0,1)\}$$\{(3,0,0),(0,0,1)\}$
- [ ] $\{(3,0,1)\}$$\{(3,0,1)\}$
- [ ] $\{(3,0,1),(0,0,1)\}$$\{(3,0,1),(0,0,1)\}$
- [ ] $\{(0,0,1)\}$$\{(0,0,1)\}$
- [ ] $\{(9,25,0),(0,0,1)\}$$\{(9,25,0),(0,0,1)\}$
- [ ] $\{(25,9,0),(0,0,1)\}$$\{(25,9,0),(0,0,1)\}$
- [ ] $\{(9,0,0),(0,25,0),(0,0,1)\}$$\{(9,0,0),(0,25,0),(0,0,1)\}$
- [ ] $\{(25,0,0),(0,9,0),(0,0,1)\}$$\{(25,0,0),(0,9,0),(0,0,1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(3,0,0),(0,0,1)\}$$\{(3,0,0),(0,0,1)\}$

$\{(3,0,1)\}$$\{(3,0,1)\}$

$\{(3,0,1),(0,0,1)\}$$\{(3,0,1),(0,0,1)\}$

$\{(0,0,1)\}$$\{(0,0,1)\}$

$\{(9,25,0),(0,0,1)\}$$\{(9,25,0),(0,0,1)\}$

$\{(9,0,0),(0,25,0),(0,0,1)\}$$\{(9,0,0),(0,25,0),(0,0,1)\}$

Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Check Answers